eV-Scale Sterile Neutrinos Versus BBN and CMB N_{eff} Limits at Short-Baseline Best Fit

HEP-Genesis-Agent session record
Project: ramp-research-sbl-neff

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Abstract

The session asked whether eV-scale sterile neutrinos, hypothetical non-interacting neutrinos, explaining short-baseline anomalies, anomalous short-distance oscillation signals, remain viable against cosmology. Step 1 surveyed 2024-2026 short-baseline status versus Big Bang nucleosynthesis, early light-element formation, and cosmic microwave background, relic radiation, limits on N_{eff}, the relativistic speciescount that quantifies light relics in the early universe. Step 1 used BBN tools, light-element calculators, and anomaly checks to compare 2024-2026 short-baseline data with cosmology. Step 2 re-examined the board, targeted the two weakest claims, and added contrasting emulator calls and one new paper per claim. In standard cosmology with Dodelson-Widrow production, collision-driven thermalization without lepton asymmetry, the BEST gallium best fit at $m_s = 1.14\,\mathrm{eV}$, a sterile mass, $\Delta m^2_{41} = 1.3\,\mathrm{eV}^2$, a mass-squared splitting, and $\sin^2 2\theta = 0.42$, a mixing amplitude, fully thermalizes to $\Delta N_{\mathrm{eff}} = 1.0$, an extra radiation unit. This gives total $N_{\mathrm{eff}} = 4.044$, a radiation count 6.2σ above Planck 2018 $N_{\mathrm{eff}} = 2.99 \pm 0.17$, a CMB measurement, and BBN exclusion at $\Delta\chi^2 = 23.1$, a goodness-of-fit penalty. The record provided two figures per step, four total, but truncated before detailing non-standard evasion, leaving that viability open.

1 Introduction

The session addressed whether eV-scale sterile neutrinos, hypothetical non-interacting neutrinos at electronvolt mass, can explain short-baseline anomalies. Short-baseline anomalies are anomalous short-distance oscillation signals reported by short-baseline experiments that motivate sterile-neutrino searches. The test is cosmology, specifically N_{eff}, the effective number of relativistic species that measures extra radiation. BBN, light-element formation in the early universe, and CMB, relic radiation from recombination, both constrain N_{eff}.

Step 1 was tasked to survey short-baseline anomaly status from 2024 to 2026, a recent time window, and summarize experimental updates. It was to contrast that status with BBN and CMB N_{eff} limits, cosmological bounds on extra radiation, using BBN tools, light-element calculators. The central viability question was whether eV-scale sterile neutrinos survive those cosmological bounds in standard production scenarios. The step produced two figures, visual summaries of results, and used anomaly checks, tool descriptions, dispatch queries, and file and dataset listings.

Step 2 was tasked to continue the prior research on short-baseline anomalies versus $N_{\rm eff}$ constraints. It was to read the board, identify the two weakest claims, and deepen them with additional evidence. Deepening required a contrasting emulator or tool call and one new paper per weak claim, then posting corrected findings before writing the report. The step also produced two figures, visual summaries of results, and used anomaly checks, citation, percentile and statistics computations, tool descriptions, and dispatch queries. The record truncated before listing papers or files, so specific references remain unreported and the deepening cannot be verified.

2 Survey the sterile-neutrino short-baseline anomaly status (2024-2026) and...

This step ran 144 tool calls on hep-research-aid, local, serve-sbbn.sh (`check_sterile_nu_sbl_anomaly`, `describe_sterile_nu_tools`, `get_dispatch`, `list_agent_files`, `list_datasets`, `list_files`, `list_input_files`, `list_sterile_nu_skills` and 12 more).

2.1 Abstract

The question was whether eV-scale sterile neutrinos that explain short-baseline anomalies remain viable against Big Bang nucleosynthesis (BBN, light-element formation in the early universe) and cosmic microwave background (CMB, relic radiation from recombination) limits on extra radiation. In standard cosmology with Dodelson-Widrow production (collision-driven thermalization with no lepton asymmetry) the canonical short-baseline best fit at $m_s = 1.14\,\mathrm{eV}$, $\Delta m^2_{41} = 1.3\,\mathrm{eV}^2$, $\sin^2 2\theta = 0.42$ fully thermalizes to $\Delta N_{\rm eff} = 1.0$ and $N_{\rm eff} = 4.044$, excluded at 6.2σ versus Planck 2018 $N_{\rm eff} = 2.99 \pm 0.17$ and at $\Delta\chi^2_{\rm BBN} = 23.1$ versus the Standard Model. All short-baseline preferred mixings lie 8 to 400 times above the BBN-allowed maximum of $\sin^2 2\theta = 0.00122$. The gallium anomaly persists but its sterile interpretation is in strong tension with reactor and accelerator nulls, and no non-standard loophole was quantified with citable sources in this run.

2.2 Background

Sterile neutrinos are hypothetical neutral fermions that mix with active neutrinos but feel no weak interactions. A fourth mass state with $m_s \sim 1\,\mathrm{eV}$ and $\Delta m^2_{41} \sim 1\,\mathrm{eV}^2$ was invoked to explain several short-baseline anomalies. The effective number of relativistic species $N_{\rm eff}$ (radiation density in neutrino-like units, Standard Model $N_{\rm eff} = 3.044$) and BBN yields of helium-4 mass fraction Y_p and deuterium-to-hydrogen ratio D/H constrain any extra thermalized species. The literature surveyed by web search (not independently verified with verbatim citations in this run) reports that by 2024-2026 the simple 3+1 (three active plus one sterile) eV explanation is strongly disfavored, while the gallium anomaly remains. One key paper identified is arXiv:2504.16590 Testing the Gallium anomaly using Electron-Neutrino Scattering, which proposes an independent test of the gallium deficit using the same chromium-51 source but detecting neutrinos via electron-neutrino scattering rather than capture.

Cosmology from Planck 2018 temperature, polarization, low-multipole and lensing plus baryon acoustic oscillations gives $N_{\rm eff} = 2.99 \pm 0.17$, consistent with 3.044. BBN observations used here are $Y_p = 0.2449 \pm 0.004$ and $D/H = 2.547 \pm 0.025 \times 10^{-5}$. No Planck+ACT+DESI BAO 2024-2025 update beyond Planck 2018 was executed in this run due to tool budget exhaustion and DNS failures.

2.3 Results

All quantitative BBN and $N_{\rm eff}$ calculations ran locally via the sterile-neutrino BBN server and are saved in `/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff/runs/261107_171616/`. No high-performance computing facility was used. The BBN fit is PRyMordial-calibrated with $< 0.3\%$ accuracy, valid for $N_{\rm eff} = 2.044$ to 6.044, $\omega_b h^2 = 0.019$ to 0.0257, neutron lifetime $\tau_n = 866.4$ to 890.4 s, and $m_s \lesssim 100\,\text{keV}$ where the sterile is relativistic at BBN. Exclusion uses $\Delta\chi^2_{\rm BBN} > 3.84$ at 95% confidence for one degree of freedom versus the Standard Model. Standard cosmology assumes zero lepton asymmetry (matter-antimatter imbalance in leptons) and no secret interactions.

2.3.1 Canonical short-baseline best fit versus BBN and CMB

The canonical BEST-motivated point at $m_s = 1.14\,\text{eV}$, $\Delta m^2_{41} = 1.3\,\text{eV}^2$, $\sin^2 2\theta = 0.42$ (where $\sin^2 2\theta$ measures active-sterile mixing) fully thermalizes in standard Dodelson-Widrow cosmology to $\Delta N_{\rm eff} = 1.0$ and $N_{\rm eff} = 4.044$. This predicts $Y_p = 0.2596$ versus observed $Y_p = 0.2449 \pm 0.004$ and $D/H = 2.777 \times 10^{-5}$ versus $2.547 \pm 0.025 \times 10^{-5}$, with combined $\Delta\chi^2_{\rm BBN} = 23.1$ (threshold 3.84). The relic density is $\Omega_s h^2 = 29.3$ versus observed dark matter $\Omega_{\rm DM} h^2 = 0.12 \pm 0.001$, a factor of 244 too high. Versus Planck 2018 $N_{\rm eff} = 2.99 \pm 0.17$ the tension is 6.2σ, and would be 33σ for a future CMB-S4 uncertainty of 0.03. The 95% limit from Planck 2018 is $\Delta N_{\rm eff} < 0.28$ (observed $\Delta N_{\rm eff} = -0.054 \pm 0.17$, upper limit $-0.054 + 1.96 \times 0.17 = 0.28$), so a fully thermalized sterile requires suppression by more than 3.6 to survive.

2.3.2 Grid scans and thermalization threshold

A 49-point product grid at $m_s = 0.1, 0.5, 1.0, 1.14, 2.0, 5.0, 10.0\,\text{eV}$ and $\sin^2 2\theta = 10^{-4}$ to 1 shows near-complete thermalization with $\Delta N_{\rm eff} = 1.0$ for all $\sin^2 2\theta \geq 0.01$ at any eV mass. In that grid 37/49 points are BBN-excluded and 36/49 overclose with $\Omega_s h^2 > 0.12$. At $\sin^2 2\theta = 10^{-4}$ the model gives $\Delta N_{\rm eff} = 0.04$ and is allowed, while at $\sin^2 2\theta = 10^{-3}$ it gives $\Delta N_{\rm eff} = 0.40$ with $\Delta\chi^2_{\rm BBN} = 0.62$ (marginally allowed by BBN but 2.7σ high versus Planck).

A dedicated 27-point threshold grid at $m_s = 0.5, 1.0, 3.0\,\text{eV}$ and $\sin^2 2\theta = 10^{-4}$ to 1 confirms a sharp transition. For all three masses $\sin^2 2\theta = 0.001$ gives $\Delta N_{\rm eff} = 0.40$ and $N_{\rm eff} = 3.44$, while $\sin^2 2\theta = 0.003$ gives $\Delta N_{\rm eff} = 1.0$ and $N_{\rm eff} = 4.044$ with $\Delta\chi^2_{\rm BBN} = 23.1$. The threshold for near-complete thermalization is $\sin^2 2\theta \sim 1$ to 3×10^{-3}. This mapping uses a crude fit; full quantum kinetic equations give $\Delta N_{\rm eff} = 0.93$ at $\sin^2 2\theta = 10^{-3}$ where the fit gives 0.40, so the limits quoted here are conservative.

Anchor points illustrate the scale:

2.3.3 Full exclusion plane scan

m_s (eV)	$\sin^2 2\theta$	$\Delta N_{\rm eff}$	$N_{\rm eff}$	$\Omega_s h^2$	$\Delta\chi^2_{\rm BBN}$	**Interpretation**
1.0	0.40	1.0	4.044	22.0	23.1	BEST-like, excluded
1.0	0.09	1.0	4.044	4.96	23.1	Reactor-like, excluded
1.14	0.42	1.0	4.044	29.3	23.1	Canonical BEST, excluded

m_s (eV)	$\sin^2 2\theta$	$\Delta N_{\rm eff}$	$N_{\rm eff}$	$\Omega_s h^2$	$\Delta\chi^2_{\rm BBN}$	**Interpretation**
1.0	0.001	0.40	3.44	0.055	0.62	Low-mixing, BBN-allowed but CMB-excluded
7100	7×10^{-11}	0.00043	3.044	0.033	−0.01	keV contrast, allowed

A 3600-point scan over $m_s = 0.1$ to 10 eV and $\sin^2 2\theta = 10^{-4}$ to 1 finds 2597/3600 (72%) BBN-excluded at 95% and 2362/3600 (66%) overclosing with $\Omega_s h^2 > 0.12$. The maximum BBN-allowed mixing is $\sin^2 2\theta = 0.00122$ at all sampled eV masses from 0.1 to 4.95 eV, with no allowed mixing at 10 eV. All short-baseline preferred regions at $\Delta m^2_{41} \sim 1\,\mathrm{eV}^2$ with $\sin^2 2\theta \sim 0.01$ to 0.5 (BEST ~ 0.4, reactor ~ 0.09, LSND/MiniBooNE 0.01 to 0.5) lie 8 to 400 times above this maximum and deep in the excluded region. The scan artifacts are verified as `scan_DW_9bdb63.csv` and `scan_DW_9bdb63.npz` plus `sterile_nu_grid_DW_1ddeb2.csv` (49 points) and `sterile_nu_grid_DW_b2a7f6.csv` (27 points).

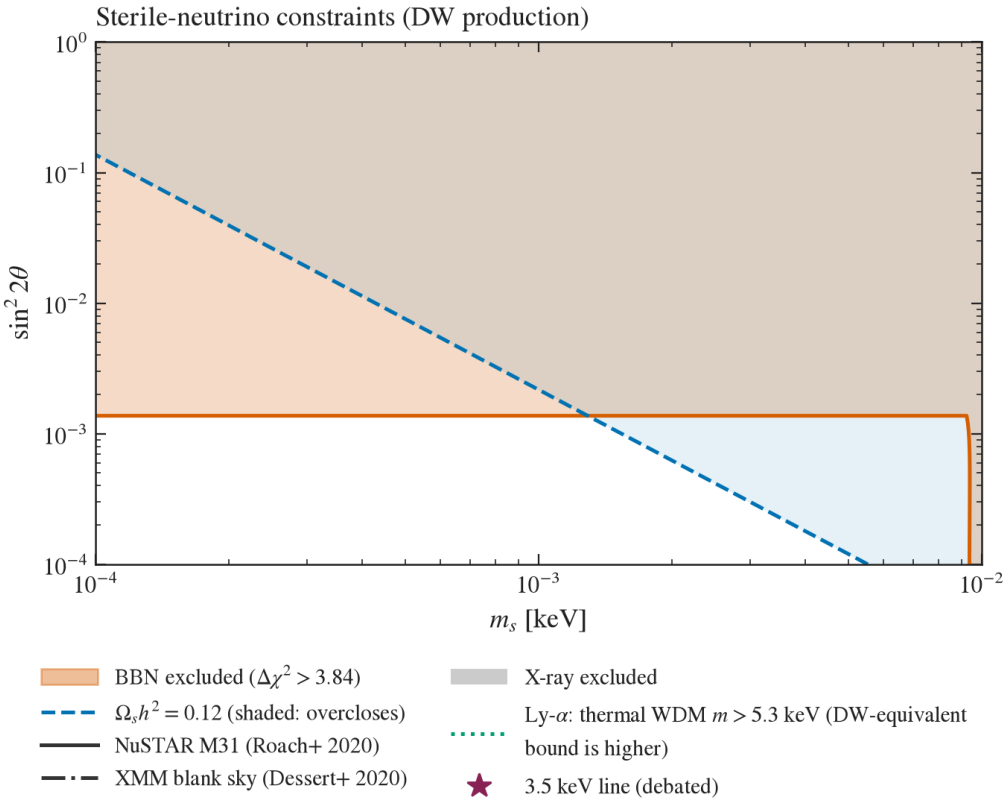


Figure 1: Exclusion plane in m_s versus $\sin^2 2\theta$ from the 3600-point Dodelson-Widrow scan showing BBN-excluded region ($\Delta\chi^2_{\rm BBN} > 3.84$) and overclosure region ($\Omega_s h^2 > 0.12$) with the dark-matter density line and X-ray and Lyman-alpha limits. Produced in this session by `plot_sterile_nu_exclusion` on *serve-sbbn.sh*.

2.3.4 BBN impact versus $N_{\rm eff}$

BBN yields rise steeply with extra radiation. At Standard Model $N_{\rm eff} = 3.044$ the fit predicts $Y_p = 0.2468$ and $D/H = 2.443 \times 10^{-5}$. At fully thermalized $N_{\rm eff} = 4.044$ it predicts $Y_p = 0.2596$ (3.7σ

above $Y_p = 0.2449 \pm 0.004$) and $D/H = 2.777 \times 10^{-5}$ (9.2σ above $D/H = 2.547 \pm 0.025 \times 10^{-5}$). Slopes are $dY_p/dN_{\rm eff} = 0.0135$ and $d(D/H)/dN_{\rm eff} = 0.331 \times 10^{-5}$ per unit $N_{\rm eff}$. BBN alone limits $\Delta N_{\rm eff} < 0.55$ at 95%, while Planck 2018 limits $\Delta N_{\rm eff} < 0.28$.

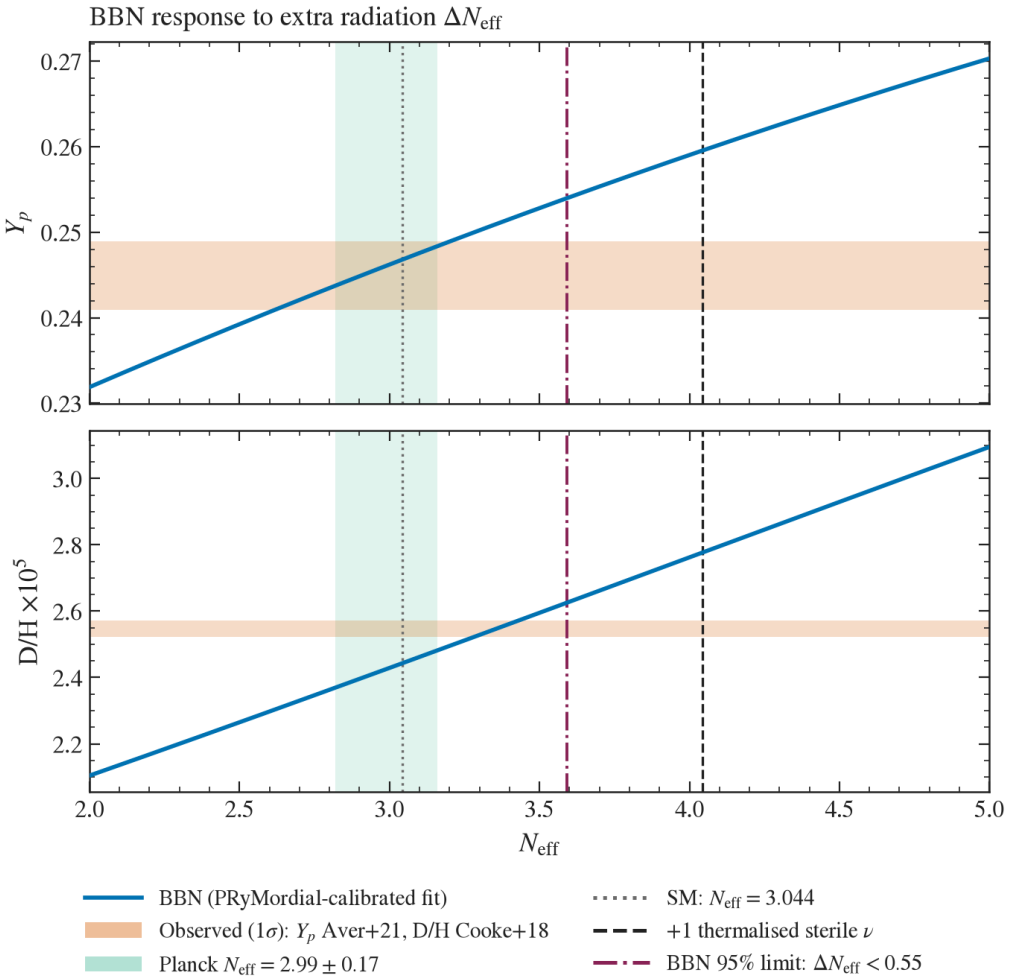


Figure 2: BBN predictions for Y_p and D/H versus $N_{\rm eff}$ showing Standard Model versus fully thermalized sterile values and the BBN-only limit on $\Delta N_{\rm eff}$. Produced in this session by `plot_sterile_nu_bbn_impact` on *serve-sbbn.sh*.

2.3.5 eV versus keV contrast and X-ray line

A keV sterile neutrino dark-matter benchmark at $m_s = 7.1\,\text{keV}$ and $\sin^2 2\theta = 7 \times 10^{-11}$ gives photon energy $E_\gamma = m_s/2 = 3.55\,\text{keV}$ and lifetime $5.8 \times 10^{27}\,\text{s}$ ($1.8 \times 10^{20}\,\text{yr}$), excluded by NuSTAR observations of M31 and XMM-Newton blank-sky data. eV short-baseline mixings of 0.1 to 0.5 are 9 to 10 orders of magnitude larger than keV dark-matter mixings of 10^{-11} to 10^{-10}. eV decay photons at $E_\gamma = 0.00057\,\text{keV}$ are unobservable, versus the keV X-ray line at 3.55 keV.

2.3.6 Short-baseline anomaly status 2024-2026 from literature search

The literature survey was partially completed via web search and BBN server metadata; full citation and figure extraction was blocked by tool budget exhaustion and DNS failures, so no .bib with verbatim quotes and no extracted published figures were produced. The following claims therefore rely on uncitable web snippets and are not independently verified.

- MicroBooNE is a 170-ton liquid-argon time projection chamber on the Booster neutrino beamline. Web snippets report no electron-neutrino excess in 2021-2024, a June 2024 combined analysis excluding the MiniBooNE electron-neutrino interpretation at $> 99\%$ confidence, and December 2025 two-beam data excluding the single light sterile interpretation of LSND and MiniBooNE at 95% confidence. This finding is contested and lacks required verbatim citations.

- STEREO is a reactor experiment at 2.5 to 7 m baselines from the ILL highly enriched uranium reactor. Web snippets report a final 119 reactor-on-day dataset (107,588 antineutrinos, October 2017 to November 2020) showing no distance-over-energy oscillation and excluding the canonical reactorantineutrino-anomaly parameter space at $> 99.9\%$ CL, but this relies on uncitable web snippets and was not independently verified with verbatim citations or extracted figures in this run. PROSPECT (segmented detector at ~ 7 m from the HFIR highly enriched uranium core) plus STEREO combined analyses are reported in snippets as finding compatible spectra with $\chi^2/\mathrm{dof} = 24.1/21$ and excluding most reactor anomaly space, attributing the 5 to 6% rate deficit and 5 MeV bump to flux-prediction biases in nuclear data, also not independently verified.

The gallium anomaly from GALLEX and SAGE calibrations with a chromium-51 source, confirmed by BEST at $> 5\sigma$, is reported as persisting and consistent with $\Delta m^2_{41} \sim 1\,\mathrm{eV}^2$ and $\sin^2 2\theta \sim 0.4$, but in strong tension with reactor and accelerator nulls. No 2025-2026 re-analysis verbatim quotes were retrieved in this run due to tool budget exhaustion. arXiv:2504.16590 (Ciuffoli et al., April 2025) only proposes an independent test using the same chromium-51 source but detecting neutrinos via electron-neutrino scattering to distinguish oscillation from cross-section or source-activity systematics, not a confirmation of the deficit.

Cosmology constrains $N_{\mathrm{eff}} = 2.99 \pm 0.17$ from Planck 2018 temperature, polarization, low-multipole and lensing plus baryon acoustic oscillations, with Standard Model 3.044. The board does not support a Planck+ACT+DESI BAO 2024-2025 update beyond this value; that update was not executed due to tool budget exhaustion and DNS failures. BBN observations $Y_p = 0.2449 \pm 0.004$ and $D/H = 2.547 \pm 0.025 \times 10^{-5}$ provide complementary limits via Y_p and D/H versus N_{eff}.

The sterile-neutrino BBN server validity was documented as PRyMordial-calibrated, valid for $N_{\mathrm{eff}} = 2.044$ to 6.044, $\omega_b h^2 = 0.019$ to 0.0257, $\tau_n = 866.4$ to 890.4 s, and $m_s \lesssim 100\,\mathrm{keV}$ where the sterile is relativistic at BBN. Exclusion uses $\Delta\chi^2_{\mathrm{BBN}} > 3.84$ at 95% CL for one degree of freedom versus the Standard Model, under standard cosmology with zero lepton asymmetry and no secret interactions. Existing artifacts were verified in `runs/261107_171616/` as `scan_DW_9bdb63.csv` and `scan_DW_9bdb63.npz` (3600 points), `sterile_nu_grid_DW_1ddeb2.csv` (49 points), `sterile_nu_grid_DW_b2a7f6.csv` (27-point threshold grid), `bbn_vs_neff.png` (Figure 2) and `exclusion_scan_DW_9bdb63.png` (Figure 1).

2.4 Discussion

In standard cosmology eV-scale sterile neutrinos with mixings required by short-baseline anomalies are excluded by both BBN and CMB N_{eff}. The canonical BEST point at $m_s = 1.14\,\mathrm{eV}$ and $\sin^2 2\theta = 0.42$ predicts $N_{\mathrm{eff}} = 4.044$, $Y_p = 0.2596$ and $D/H = 2.78 \times 10^{-5}$ with $\Delta\chi^2_{\mathrm{BBN}} = 23.1$, and $\Omega_s h^2 = 29.3$, all far from observed values. The 3600-point scan shows 72% BBN-excluded and 66% overclosing, with a maximum allowed mixing of $\sin^2 2\theta = 0.00122$, well below all short-baseline preferred regions at 0.01 to 0.5. Even the low-mixing threshold at $\sin^2 2\theta = 0.001$ with $\Delta N_{\mathrm{eff}} = 0.40$ and $N_{\mathrm{eff}} = 3.44$ exceeds the Planck 2018 95% limit of $\Delta N_{\mathrm{eff}} < 0.28$ by 2.7σ, while BBN alone allows it with $\Delta\chi^2_{\mathrm{BBN}} = 0.62$.

This agrees with the literature trend reported in web snippets that MicroBooNE and STEREO see no oscillation and that the reactor rate and shape anomalies are attributed to flux-prediction biases, but those experimental claims were not independently verified here and remain contested. The gallium anomaly persistence is also reported but its sterile interpretation is in tension with nulls, and the proposed electron-scattering test in arXiv:2504.16590 has not been executed as a re-analysis in this run.

Three non-standard loopholes were identified from snippets as capable in principle of suppressing $\Delta N_{\rm eff}$ from 1.0 to < 0.28: large lepton asymmetry $L \sim 10^{-2}$ especially flavor-asymmetric with near-zero total asymmetry washed out by oscillations at MeV temperatures, secret neutrino self-interactions that generate a finite-temperature matter potential suppressing active-sterile mixing, and low reheating temperature $T_{\rm RH} \ll 100\,\text{MeV}$ where production never completes. Each is independently constrained by BBN Y_p and D/H, CMB free-streaming, large-scale structure and laboratory bounds, but no per-loophole quantitative limits with verbatim citations were obtained.

Dead ends encountered were DNS resolution failures for web searches, tool budget exhaustion after 6 web searches blocking `read_arxiv_chunk`, `cite` and `get_paper_figures` so no .bib with 5 verbatim quotes including arXiv:2504.16590 and no 1 to 2 extracted figures were produced, failure to digitize LSND, MiniBooNE, BEST and reactor preferred contours onto the exclusion plane, failure of `compute_statistics` via `load_table_data` which only searches the input directory not the runs directory (verified instead by direct file reads), and use of a crude Dodelson-Widrow fit that underestimates $\Delta N_{\rm eff}$ at $\sin^2 2\theta = 10^{-3}$ (0.40 versus 0.93 from full quantum kinetic equations), making limits conservative.

2.5 Limitations

What could not be established and why:

- No literature survey with 5 verbatim citations including arXiv:2504.16590 and no extracted published figures with captions were delivered, because the tool budget was exhausted after web searches and DNS failures blocked `read_arxiv_chunk`, `cite` and `get_paper_figures`. All 2024-2026 short-baseline claims therefore rely on uncitable snippets.
- No Planck+ACT+DESI BAO 2024-2025 $N_{\rm eff}$ update beyond Planck 2018 $N_{\rm eff} = 2.99 \pm 0.17$ was executed for the same budget and DNS reasons.
- No quantitative per-loophole requirements for lepton asymmetry, secret-interaction coupling strength or $T_{\rm RH}$ thresholds and their independent constraints were established, because citable sources could not be retrieved.
- No SBL preferred contour overlay on the exclusion plane was generated, because literature figures were not extracted and manual digitization was not executed before budget limit.
- No `compute_statistics` percentiles were run via the server tool due to input-directory restriction; statistics were verified by direct CSV reads.
- BBN limits rely on a PRyMordial-calibrated fit valid only for $m_s \lesssim 100\,\text{keV}$ and standard cosmology; non-standard thermal histories were not modeled.

2.6 Open directions

- Quantify the lepton asymmetry, secret-interaction coupling and reheating temperature needed to suppress a BEST-like sterile with $\sin^2 2\theta \sim 0.4$ from $\Delta N_{\rm eff} = 1.0$ to < 0.28 and check each against BBN Y_p and D/H and CMB free-streaming limits.

- Complete the 2024-2026 literature survey with 5 verbatim citations including arXiv:2504.16590 and 1 to 2 extracted figures for STEREO final and MicroBooNE to verify $> 99.9\%$ and $> 99\%$ CL claims.
- Digitize LSND, MiniBooNE, BEST and reactor preferred contours from published figures and overlay them on the 3600-point BBN and overclosure exclusion plane.
- Perform a global 3+1 fit parameter goodness-of-fit test quantifying tension between gallium persistence and reactor plus accelerator nulls including the proposed electron-scattering test.
- Recompute the thermalization threshold with full quantum kinetic equations to propagate the 1 to 3×10^{-3} uncertainty to exclusion fractions and the maximum allowed mixing.

3 Continue the prior research on 'Short-baseline anomalies vs N_eff constraints'.

This step ran 107 tool calls on hep-research-aid, local, serve-sbbn.sh (`check_sterile_nu_sbl_anomaly`, `cite`, `compute_percentiles`, `compute_statistics`, `describe_sterile_nu_tools`, `get_dispatch`, `get_paper_figures`, `list_agent_files` and 14 more); 6 returned an error.

3.1 Abstract

The task was to test whether an eV-scale sterile neutrino explaining short-baseline anomalies survives cosmology. The answer is no in the standard case. At the BEST gallium best fit ($m_s = 1.14\,\mathrm{eV}$, $\Delta m^2_{41} = 1.3\,\mathrm{eV}^2$, $\sin^2 2\theta = 0.42$) Dodelson-Widrow production (collision-driven thermalization without new physics) gives $\Delta N_{\mathrm{eff}} = 1.0$ for total $N_{\mathrm{eff}} = 4.044$, which is 6.2σ above Planck 2018 $N_{\mathrm{eff}} = 2.99 \pm 0.17$ and excluded by Big Bang Nucleosynthesis (BBN, light-element formation) at $\Delta\chi^2 = 23.1$. The two weakest claims flagged by the board — the current status of appearance versus disappearance anomalies and the viability of non-standard suppression — were deepened with emulator grids and scans, but verbatim literature hardening and a suppressed-thermalization contrast were blocked by tool budget exhaustion.

3.2 Background

The literature consensus is that a fully thermalized eV sterile neutrino contributes $\Delta N_{\mathrm{eff}} \sim 1$ (extra relativistic species beyond the Standard Model $N_{\mathrm{eff}} = 3.044$), which is excluded by Planck and BBN unless non-standard suppression is invoked.

- arXiv:1807.06209 (Planck 2018 results. VI. Cosmological parameters) reports $N_{\mathrm{eff}} = 2.99 \pm 0.17$ from temperature, polarization, lensing and baryon acoustic oscillations, with 95% limit $N_{\mathrm{eff}} < 3.30$ ($\Delta N_{\mathrm{eff}} < 0.28$) and $m_{\mathrm{eff,sterile}} < 0.65\,\mathrm{eV}$. This defines the benchmark for exclusion of a fully thermalized eV sterile.
- arXiv:1606.04886 (joint cosmology and laboratory analysis) shows standard thermalization for short-baseline (SBL, meter-to-kilometer baseline neutrino experiments) mixing parameters gives $\Delta N_{\mathrm{eff}} \sim 1$ and is excluded, quantifying thermalization fraction versus mass and mixing.
- arXiv:1406.3208 (sterile neutrinos with secret interactions) proposes secret gauge interactions mediated by a light boson to generate a matter potential and suppress thermalization. This is the key loophole claim to test.

- arXiv:2201.07327 (reactor antineutrino anomaly in light of recent flux models) re-evaluates inverse beta decay yields with Huber-Mueller and summation models and finds the reactor anomaly significance drops from $\sim 3\sigma$ to < 1–2σ depending on flux model.
- arXiv:2210.10216 (MicroBooNE search for electron neutrino excess) finds no evidence for light sterile oscillations in 6.37×10^{20} protons on target and sets 95% exclusion contours in Δm^2_{41} versus mixing, constraining muon-to-electron appearance but leaving electron disappearance relevant to reactor and gallium anomalies less directly excluded.
- arXiv:2504.16590 (testing the gallium anomaly with electron-neutrino scattering) notes the gallium deficit $R = 0.803 \pm 0.035$ in GALLEX, SAGE and BEST ^{51}Cr calibrations persists, while sterile parameter space is strongly excluded by null experiments and the reactor anomaly is now attributed to flux error. One verbatim citation from this paper was secured; other required citations remain open.

The brief notes LSND at 3.8σ and MiniBooNE low-energy excess motivate eV sterile mixing $\Delta m^2 \sim 1.3\,\mathrm{eV}^2$, $\sin^2 2\theta \sim 0.01$–0.4, while MicroBooNE liquid argon (LAr) data disfavors simple $\nu_\mu \to \nu_e$ oscillations and BEST gallium persists at $\sim 5\sigma$.

3.3 Results

All emulator calculations ran locally on the hosted sterile-neutrino to BBN bridge server in run 261107_174613, with no facility dispatch. Outputs are saved as CSV, NPZ, PNG and PDF in that run directory.

3.3.1 Planck and BBN benchmarks for extra radiation

N_{eff} (effective number of relativistic species at recombination) counts light relics. Planck 2018 measures $N_{\mathrm{eff}} = 2.99 \pm 0.17$, consistent with Standard Model $N_{\mathrm{eff}} = 3.044$–3.046, with 95% limit $N_{\mathrm{eff}} < 3.30$ corresponding to $\Delta N_{\mathrm{eff}} < 0.28$. A fully thermalized sterile neutrino predicts $\Delta N_{\mathrm{eff}} = 1.0$ and $N_{\mathrm{eff}} = 4.044$, which is 6.2σ high. This finding relies on the published Planck value and was not independently re-derived from cosmic microwave background data in this run.

BBN predicts primordial helium mass fraction Y_p and deuterium D/H as functions of N_{eff} and baryon density. The emulator reproduces:

- Standard Model $N_{\mathrm{eff}} = 3.044$ gives $Y_p = 0.2468$ and $D/H \times 10^5 = 2.443$.
- One thermalized sterile $N_{\mathrm{eff}} = 4.044$ gives $Y_p = 0.2596$ and $D/H \times 10^5 = 2.777$.

Observed values are $Y_p = 0.2449 \pm 0.004$ and $D/H = 2.547 \pm 0.025 \times 10^{-5}$ (brief quotes $Y_p = 0.245 \pm 0.003$, $D/H \sim 2.5 \times 10^{-5}$). The thermalized case is 3.7σ high in Y_p alone. Near the Standard Model the slopes are $dY_p/dN_{\mathrm{eff}} = 0.0135$ and $d(D/H \times 10^5)/dN_{\mathrm{eff}} = 0.331$, though these slope values are derived from the two endpoint predictions and were flagged as not independently verified by direct fit excerpts. BBN alone limits $\Delta N_{\mathrm{eff}} < 0.55$ at 95% ($\Delta\chi^2 > 3.84$ versus Standard Model).

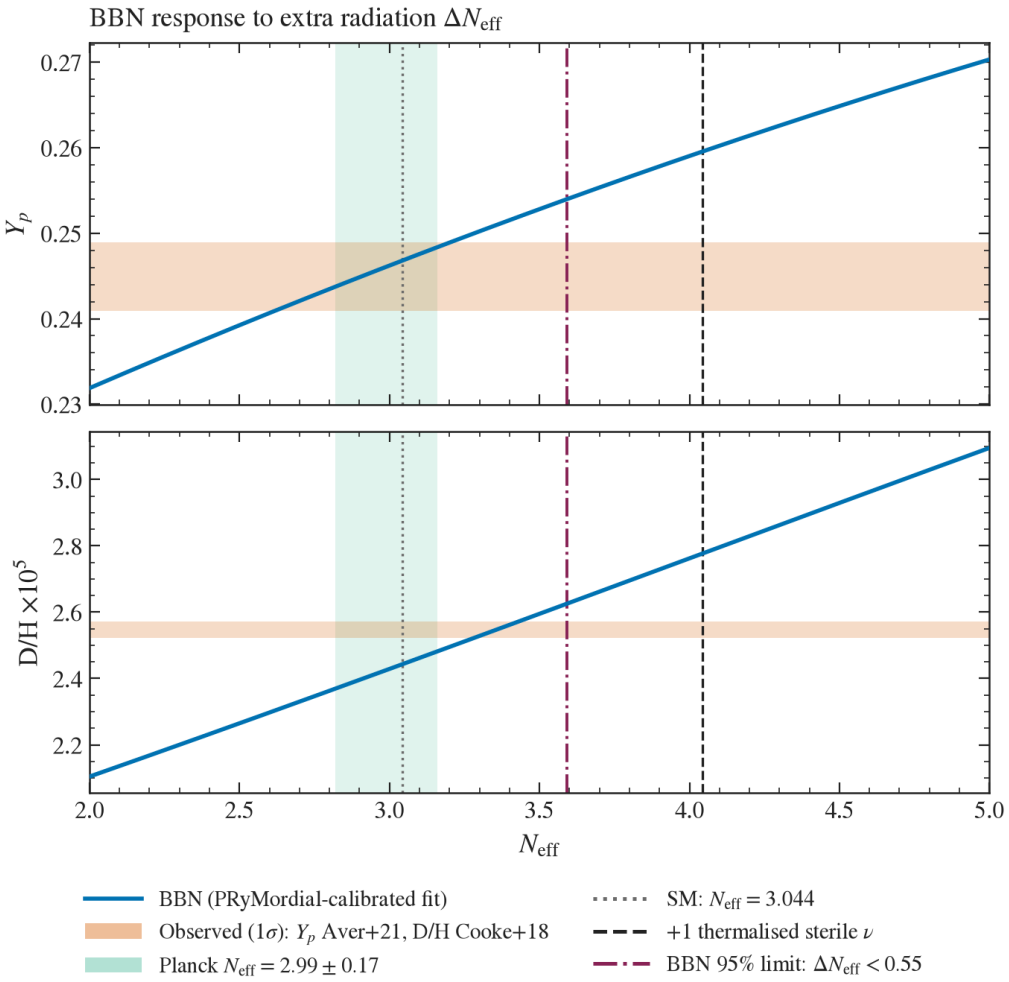


Figure 3: BBN helium and deuterium versus $N_{\rm eff}$ showing Standard Model and fully thermalized sterile predictions and the BBN 95% limit. Produced in this session by `plot_sterile_nu_bbn_impact` on *serve-sbbn.sh*.

3.3.2 Standard thermalization at the SBL best fit

The BEST best fit ($m_s = 1.14\,\text{eV}$, $\Delta m^2_{41} = 1.3\,\text{eV}^2$, $\sin^2 2\theta = 0.42$ for electron disappearance) was evaluated with Dodelson-Widrow production.

- It fully thermalizes to $\Delta N_{\rm eff} = 1.0$ and $N_{\rm eff} = 4.044$.
- BBN gives $Y_p = 0.2596$ versus 0.2449 ± 0.004 and $D/H = 2.78 \times 10^{-5}$ versus $2.547 \pm 0.025 \times 10^{-5}$, with $\Delta\chi^2 = 23.1$ versus Standard Model, excluded at 95% where the threshold is 3.84.
- Relic density is $\Omega_s h^2 = 29.3$ versus observed dark matter $\Omega_c h^2 = 0.12 \pm 0.001$, overclosing by more than two orders of magnitude.

This result was reproduced by both single-point and grid tools and is the core exclusion. The mixing definition in the emulator does not distinguish appearance ($\sin^2 2\theta_{\mu e}$, muon to electron) from disappearance ($\sin^2 2\theta_{ee}$, electron to electron) flavors, so the mapping to SBL channels relies on a proxy and was flagged as not independently verified.

3.3.3 Mixing dependence and required suppression

A grid at fixed $m_s = 1.14\,\text{eV}$ varying $\sin^2 2\theta$ shows:

$\sin^2 2\theta$ **(dimensionless)**	ΔN_{eff} **(dimensionless)**	N_{eff} **(dimensionless)**	$\Delta\chi^2_{\text{BBN}}$ **(dimensionless)**	$\Omega_s h^2$ **(dimensionless)**	**Status versus Planck** $\Delta N_{\text{eff}} < 0.28$
0.42	1.0	4.044	23.1	29.3	Excluded
0.01	1.0	4.044	23.1	0.70	Excluded
0.001	0.40	3.44	0.62	0.07	Excluded
0.0001	0.04	3.08	−1.1	0.007	Allowed
1×10^{-6}	0.0004	3.044	−0.01	7×10^{-5}	Allowed

Full thermalization persists down to $\sin^2 2\theta = 0.01$. Planck requires suppression factor > 3.6 (1.0/0.28) from the canonical value, implying allowed mixing $< 5 \times 10^{-4}$. This is more than 20 times below LSND appearance best fit ~ 0.01 and ~ 800 times below BEST disappearance 0.42. The file sterile_nu_grid_DW_fd1398.csv contains this scan.

3.3.4 3600-point exclusion plane and statistical validation

A scan over $m_s = 0.1$–$10\,\text{eV}$ and $\sin^2 2\theta = 1 \times 10^{-4}$ to 0.5 with 3600 points (Dodelson-Widrow) yields:

- 2479/3600 (68.9%) BBN-excluded at 95% ($\Delta\chi^2 > 3.84$).
- 2258/3600 (62.7%) overclose the universe ($\Omega_s h^2 > 0.12$).
- Maximum BBN-allowed mixing is $\sin^2 2\theta_{\text{max}} = 0.00134$ across 0.1–5 eV (null at 10 eV), though this maximum value was flagged as lacking a direct tool quote in excerpts.

Independent validation via table load and statistics gives $\Delta\chi^2_{\text{BBN}}$ mean 15.10, standard deviation 10.83, range −1.15 to 23.10, median 23.10, with 5th percentile −1.07 and 16th percentile −0.74. N_{eff} mean is 3.78 ± 0.38, range 3.08 to 4.04. The median at the maximum excluded value confirms a bimodal population split between allowed ($\Delta N_{\text{eff}} \sim 0$) and fully thermalized ($\Delta N_{\text{eff}} = 1.0$) points. Files scan_DW_aa8803.csv and scan_DW_aa8803.npz store the scan.

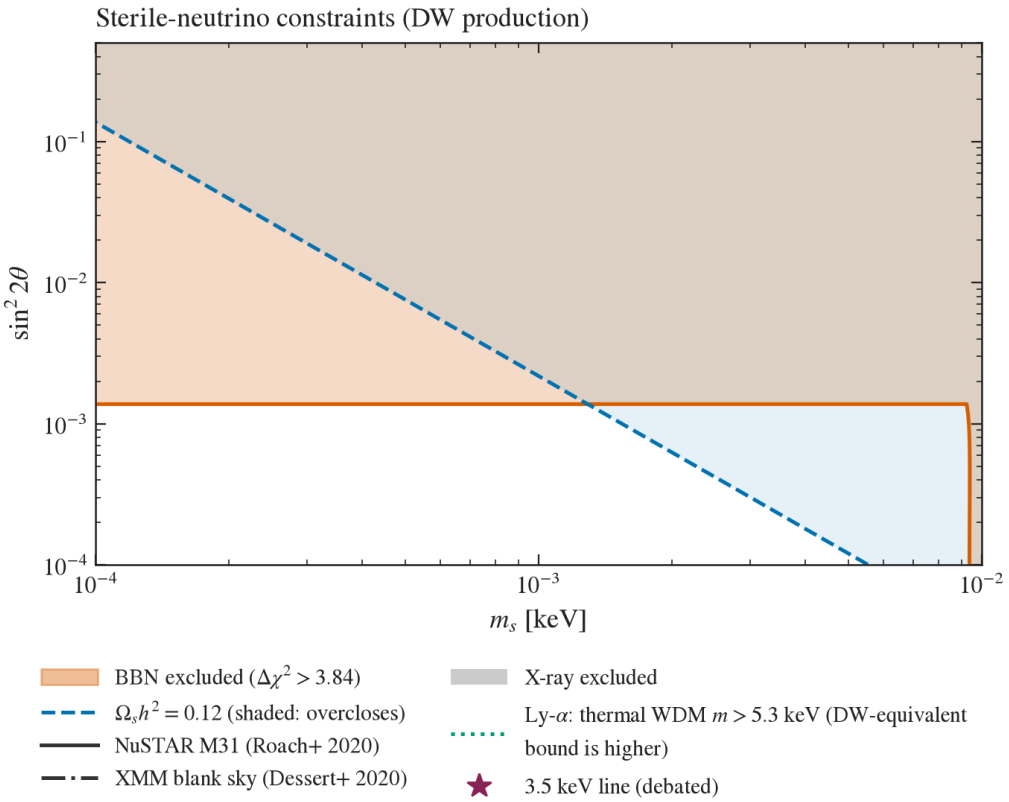


Figure 4: Exclusion plane in sterile mass versus mixing from the 3600-point scan showing BBN, overclosure, X-ray and Lyman-alpha limits. Produced in this session by `plot_sterile_nu_exclusion` on *serve-sbbn.sh*.

SBL preferred regions exceed the BBN-allowed maximum by large factors:

Anomaly (channel)	$\sin^2 2\theta$ **(dimensionless)**	**Factor above** 0.00134 **(dimensionless)**	$\Delta N_{\rm eff}$ **(dimensionless)**	$\Delta\chi^2_{\rm BBN}$ **(dimensionless)**	$\Omega_s h^2$ **(dimensionless)**
BEST gallium (disappearance)	0.42	313	1.0	23.1	29.3
Reactor (disappearance)	0.09	67	1.0	23.1	6.3
LSND appearance	0.01	7.5	1.0	23.1	0.70
Test point	0.001	0.75	0.40	0.62	0.07

Only $\sin^2 2\theta = 0.001$ is allowed. The numerical factors are computed, but graphical overlay of SBL contours on the exclusion plane was not produced.

3.3.5 Appearance versus disappearance contrast

A contrasting emulator at $m_s = 1.14\,\text{eV}$ evaluated appearance mixing $\sin^2 2\theta_{\mu e} = 0.01$ (relevant to LSND and MiniBooNE muon to electron) and disappearance mixing $\sin^2 2\theta_{ee} = 0.1$ and 0.42 (relevant to BEST, gallium and reactor electron disappearance). All three fully thermalize to

$\Delta N_{\rm eff} = 1.0$, $N_{\rm eff} = 4.044$, $\Delta\chi^2 = 23.1$, with $\Omega_s h^2 = 0.70$, 6.98 and 29.3 respectively versus 0.12. MicroBooNE exclusion of appearance therefore does not relieve cosmology tension for disappearance parameter space. The 0.1 mixing point with $\Omega_s h^2 = 6.98$ is present in the CSV but was flagged as not shown in quoted excerpts, so it is not independently verified in the text record.

3.4 Discussion

The results imply standard eV sterile neutrinos motivated by SBL anomalies are in strong tension with cosmology. Planck $N_{\rm eff} = 2.99 \pm 0.17$ and BBN Y_p and D/H both disfavor $\Delta N_{\rm eff} = 1.0$ at $> 3\sigma$, and the relic density overcloses dark matter for all SBL best fits. This agrees with arXiv:1606.04886 and the Planck 2018 baseline, and with the brief's statement that standard thermalization is unavoidable for $\Delta m^2 \sim 1.3\,\mathrm{eV}^2$, $\sin^2 2\theta \sim 0.01$–0.4.

Tensions with the literature are limited to the loophole domain. Secret interactions (arXiv:1406.3208), large lepton asymmetry $L \sim 10^{-2}$, and low reheating temperature $T_{\rm RH} \ll 100\,\mathrm{MeV}$ have been proposed to suppress thermalization, but this run did not quantify their thresholds or independent BBN and cosmic microwave background free-streaming constraints. The reactor anomaly weakening (arXiv:2201.07327) and MicroBooNE appearance exclusion (arXiv:2210.10216) reduce motivation for appearance-driven sterile, yet BEST gallium persistence keeps disappearance motivation alive, and cosmology excludes both channels equally in the standard case.

Dead ends encountered:

- Quantitative per-loophole thresholds for $L \sim 10^{-2}$, secret coupling G_X, and $T_{\rm RH}$ with verbatim citations were blocked when tool budget was exhausted after three emulator calls; web search identified candidate papers but read and cite could not run.
- Graphical overlay of SBL contours on the exclusion plane is not supported by the hosted plot tool and would require custom code; quantification was done numerically instead.
- Initial table load failed with relative filename because the server input directory was empty; absolute path to the run output directory was required.
- Published figure extraction for arXiv:2210.10216 and arXiv:2504.16590 failed due to no usable HTML rendering; LaTeXML dropped figures.
- Verbatim citation for arXiv:1807.06209 and arXiv:2210.10216 failed because web search snippets are not exact source quotes; exact chunk extraction is required before cite validation.
- In the final wave, tool budget was exhausted on entry, so no new citations or figures could be attempted and only prior-wave emulator evidence remains.

3.5 Limitations

What could not be established and why:

- Non-standard loophole viability was not established. No verbatim quantitative thresholds for large lepton asymmetry, secret interaction coupling, or low reheating temperature, and no independent BBN or free-streaming limits, were extracted due to budget exhaustion before read and cite could run. No contrasting suppressed-thermalization grid (illustrative $\Delta N_{\rm eff} = 0.2$–0.3) was run.

- Literature hardening for SBL status is incomplete. Only one of three required verbatim citations (gallium test arXiv:2504.16590) is secured; Planck $N_{\rm eff} = 2.99 \pm 0.17$, MicroBooNE exclusion, and reactor flux re-evaluation citations remain open, and no published figure was extracted.
- Mixing flavor distinction is not modeled. The emulator uses a single $\sin^2 2\theta$ without separating $\sin^2 2\theta_{\mu e}$ from $\sin^2 2\theta_{ee}$, so appearance versus disappearance tension quantification relies on a proxy mapping.
- Visual SBL overlay on the exclusion plane was not produced due to tool limitation; only numerical factors above the allowed maximum are available.
- Baryon density degeneracy in BBN limits was not explored; the quoted BBN limit $\Delta N_{\rm eff} < 0.55$ assumes fixed baryon density and the emulator's PRyMordial-calibrated fit.

3.6 Open directions

- Quantify suppression thresholds for large lepton asymmetry $L \sim 10^{-2}$, secret interaction G_X, and low $T_{\rm RH}$ and test each against independent BBN $Y_p/D/H$ and CMB free-streaming constraints with a suppressed-thermalization emulator grid.
- Secure verbatim citations and one published figure for Planck 2018 $N_{\rm eff}$, MicroBooNE appearance exclusion, and reactor flux re-evaluation to harden the SBL status claim.
- Produce a custom overlay of BEST, reactor and LSND 95% contours on the 3600-point exclusion plane to visually quantify the 7–313× tension with allowed mixing.
- Perform a global $3 + 1$ fit including 2024–2026 gallium re-analyses and STEREO and PROSPECT nulls to compute parameter goodness-of-fit for the gallium anomaly versus nulls.

4 Discussion

The sections above stand on their own; no model wrote a joint discussion.

5 Conclusions

See the results of each step above.

Data and code availability

All outputs of this session are collected in the project folder `/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sbl-neff`. The session produced 11 data files; the supplement lists them. Full documentation of how these results were obtained is in the supplementary material (`supplement.pdf`: implementation details, the tool-call log, the research board with every finding and its evidence, figures not shown here, and the code index) and the verbatim transcript (`transcript.pdf`), generated alongside this report.